

A Project Report on

Multimodal Depression Detection System using Fine-Tuned RoBERTa and Wav2Vec2

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by

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MAY 2026

CERTIFICATE

This is to certify that the REPORT entitled **Multimodal Depression Detection System using Fine-Tuned RoBERTa and Wav2Vec2** submitted by **Laraib Firoz** to the Manipal University Jaipur, in partial fulfilment for the requirement of the degree of **Bachelor of Technology** in Computer and Communication Engineering is a *bona fide* record of REPORT carried out by him under my/our supervision. The contents of this REPORT, in full or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma.

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Place: Manipal University Jaipur

May 2026

DECLARATION

I certify that

- a. The work contained in this project is original and has been done by myself and the general supervision of my supervisor.
- b. The work has not been submitted to any other institute for any degree or diploma.
- c. Whenever I have used materials (data, theoretical analysis, results) from other sources, I have given due credit to them by citing them in the text of the REPORT and giving their details in references.
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Signature of the Student

Date: 15 May 2026

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Laraib Firoz
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ABSTRACT

The most common type of mental illness is depression, but many people with depression go undiagnosed because conventional measures of depression have traditionally depended on subjective ways (i.e., interviews or self-reports) to conduct the assessment. Early identification of the symptoms of depression is essential for the effective management of depression, and current diagnostic measures used to determine depression are both limited by the inability for the assessment method to both scale and provide an objective diagnosis of depression. In response to this problem, this paper describes an automated method for the detection of the symptoms of depression using a data-driven approach based on the latest advancements in artificial intelligence (AI), specifically a method known as multimodal learning.

Utilising both text and audio (i.e., speech) that has been collected from the participants, we developed a multimodal learning method for integrating the use of two different data types using pre-trained transformer-based models, RoBERTa (for text data) and Wav2Vec2 (for audio data). Due to the potential issues associated with both overfitting and small sample size, we will only be fine-tuning the final layers of the RoBERTa and Wav2Vec2 models respectively and we will be leaving all other layers of these two models frozen when implementing this method. The features derived from each of the models will then be transformed into a common embedding space, merged together to create a complete representation of the data, and subsequently classified using machine learning techniques. Finally, the method will be trained and tested using the DAIC-WOZ dataset with three-fold stratified cross-validation for the purpose of demonstrating robustness in the presence of class imbalance.

Experimental results demonstrate that the model achieves a mean accuracy of 88.8%, an F1-score of 83.6%, and an AUC of 91.6%, indicating strong performance and generalization capability. The findings highlight the effectiveness of multimodal deep learning in mental health analysis and suggest that the proposed system can serve as a reliable tool for early depression screening, supporting clinicians with objective and scalable assessment methods.

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ABBREVIATIONS

AI	Artificial Intelligence
NLP	Natural Language Processing
ASR	Automatic Speech Recognition
AUC	Area Under Curve
ROC	Receiver Operating Characteristic
F1-score	Harmonic Mean of Precision and Recall
LLM	Large Language Model
ML	Machine Learning
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
BCE	Binary Cross Entropy
PHQ-8	Patient Health Questionnaire (8-item)
LLM	Large Language Model
ReLU	Rectified Linear Unit

NOTATIONS

x_t	Input text
x_a	Input audio signal
$y.$	Ground truth label (0: Non-depressed, 1: Depressed)
$\hat{y}$	Predicted output label
$f(\cdot)$	Classification function of the model
$T(\cdot)$	Text feature extraction function (RoBERTa)
$A(\cdot)$	Audio feature extraction function (Wav2Vec2)
z_t	Text feature vector
z_a	Audio feature vector
z	Fused feature representation
σ	Sigmoid activation function
θ	“Element of” (set membership)

NOMENCLATURE

Multimodal Learning prediction	Technique that combines multiple data types (text and audio) for improved prediction
Depression Detection	Process of identifying depressive conditions using computational methods
Transcript	Textual representation of spoken clinical interviews
Acoustic Features	Speech-related characteristics such as tone, pitch, and pauses
Feature Extraction	Process of deriving meaningful representations from raw data
Feature Fusion	Combining features from multiple modalities into a unified representation
Classification	Process of predicting whether a subject is depressed or not
Cross-Validation	Technique for evaluating model performance using multiple data splits
Class Imbalance	Unequal distribution of classes in the dataset
Overfitting	When a model performs well on training data but poorly on unseen data
Pre-trained Model	Model trained on large datasets and reused for specific tasks
Fine-Tuning	Adjusting a pre-trained model for a specific task
Dropout	Regularization technique to prevent overfitting
Early Stopping	Technique to stop training when performance stops improving

CHAPTER 1: INTRODUCTION

1. Background

Depression is a common mental health disorder that affects millions of people of all ages around the world. Even though many people are affected by depression, the majority of them go undiagnosed or misdiagnosed due to how traditional assessment methods such as clinical interviews and self-reported questionnaires rely on a person's subjective experience as well as a clinician's interpretation of their experience. Due to the thorough reliance of traditional assessment approaches, there can be inconsistencies between different assessments and consequently, there is generally a great deal of time that passes before someone is accurately diagnosed.

The rapid growth and expansion of Artificial Intelligence (AI) and Machine Learning (ML) have led to increased interest in developing automated systems that analyze mental health data. In both cases, by using Natural Language Processing (NLP) and other speech processing techniques, it is possible to analyze the text or speech of that person to provide a better understanding of that person's emotional or psychological state. In this sense, because depression can manifest itself in both how an individual expresses themselves (the text they use) and how an individual speaks (the manner or way in which they speak), multimodal learning, which refers to the use of multiple types of learning data, has become an emerging and potent way to enhance the accuracy of detecting depression among individuals.

This project will utilize multimodal deep learning techniques to build an intelligent system for detecting depression. By integrating both textual and acoustic data into a single assessment via advanced transformer models, the system will provide a more thorough and objective assessment than can be obtained through traditional assessment approaches.

2. Motivation

The Rise of Mental Health Issues, More Specifically Depression, is due to Academic Pressure, Changing Lifestyles, and Social Issues; These Issues Cause Many People to Not Seek Professional Help; For Example, Stigma; Awareness; and Access (to Mental Health Resources) are Some of the Reasons for Not Seeking Professional Help. As a Result, There is a Need for Accessible and Non-Intrusive Tools to Identify and Monitor Early Mental Health Problems.

Most Current Automated Detection Approaches of Depression rely on One Modality; this Reduces All the Different Methods of Expression of an Emotion; For Example, Through Text and/or Speech, As Well as Vocal Tone, Facial Expression, Body Language, Etc. Additionally,

most Detection Models Overfit and are Not Robust due to Limited, Incomplete, and/or Unbalanced Datasets. These Challenges Indicate a Need for A More Reliable and Scalable Solution.

Therefore, the Purpose of This Project is to Create a Multimodal Detection System that Combines Textual Features with Acoustic Features to Improve Detection Performance and to Create Robustness Through Controlled Fine Tuning and Evaluation Methods; the Detection System is Intended to Provide a Tool for Both Clinicians and Individuals to Track Early Signs of Depression in A Data Driven and Objective Way.

3. Problem Statement

Traditional methods for diagnosing depression rely on subjective evaluations, which can lead to inconsistent results and delayed intervention. Additionally, existing automated systems often fail to effectively combine multiple sources of information, resulting in reduced accuracy and generalization.

Therefore, the problem addressed in this project is:

“To design and develop an automated multimodal system that can accurately detect depression by analyzing both textual and acoustic features, ensuring robustness, scalability, and reliable performance under real-world conditions.”

4. Objectives

The primary goals of this project include:

- Constructing a multimodal deep learning model to detect depression with textual and acoustic data.
- Implementing transformer-based models (RoBERTa and Wav2Vec2) for the extraction of features from both transcript and audio data.
- Developing an effective feature fusion method to integrate both modalities into one cohesive representation.
- Utilizing controlled fine-tuning techniques to avoid overfitting and increase the model's ability to generalize.
- Assessing the performance of the system using performance measures, including accuracy, F1 score, and area under the curve.
- Providing rigorous evaluation using stratified cross-validation with a real data set.

5. Scope of the Project

Using clinical interview data, this project will create a multimodal depression detection system. The system will analyze how text transcripts corresponded to their related audio recordings and use the results to predict the likelihood of an individual being depressed or not. The main components of the project include data preprocessing, model design, training of the model, and evaluation of the model.

The project is limited in scope as it only uses the DAIC-WOZ dataset and does not explore real-time applications or integration of the model into clinical environments. The system performs well, but the intent is to use it as a way to support 'professional diagnosis' rather than to replace them. Future project enhancements are expected to include real-time applications, additional data sets that are of greater size than the DAIC-WOZ dataset, and integration with the healthcare system to allow for practical application.

CHAPTER 2: SYSTEM REQUIREMENTS

1. Hardware Requirements

The development and execution of the proposed multimodal depression detection system require moderate computational resources, especially due to the use of deep learning models for text and audio processing.

- **Processor:** Intel Core i5 or higher (recommended: i7 or equivalent)
- **RAM:** Minimum 8 GB (16 GB recommended for smooth training)
- **Storage:** At least 20 GB free space for dataset, model weights, and logs
- **GPU (Optional but Recommended):** NVIDIA GPU with CUDA support for faster training and inference
- **System:** Windows 10/11, macOS, or Linux
- **Audio Support:** Standard audio processing capability for handling waveform data

2. Software Requirements

1. Programming Environment

- Language: Python 3.8 or above
- IDE/Tools: VS Code, Jupyter Notebook, or Google Colab
- Version Control: Git/GitHub

2. Machine Learning & Deep Learning Libraries

- PyTorch: For model implementation and training
- HuggingFace Transformers: For pre-trained models such as RoBERTa and Wav2Vec2
- scikit-learn: For evaluation metrics and preprocessing utilities
- NumPy & Pandas: For data handling and manipulation

3. Audio Processing Libraries

- Librosa / Torchaudio: For loading, preprocessing, and feature extraction from audio signals

4. Visualization & Analysis Tools

- Matplotlib / Seaborn: For plotting training curves, ROC curves, and performance metrics

5. Dataset

- DAIC-WOZ Dataset: Clinical interview dataset containing transcripts, audio recordings, and depression labels

3. Functional Requirements

The system is expected to perform the following core functions:

Data Preprocessing

- Load and clean textual transcripts (removal of noise, tokenization, normalization)
- Process raw audio signals (resampling, padding, trimming)
- Handle missing or corrupted data samples

Feature Extraction

- Extract contextual embeddings from text using RoBERTa
- Extract acoustic representations from audio using Wav2Vec2

Multimodal Fusion

- Project both textual and acoustic features into a common feature space
- Combine features to form a unified representation for classification

Model Training

- Train the model using stratified cross-validation
- Apply techniques such as class weighting, dropout, and gradient clipping

Prediction

- Classify input data into depressed or non-depressed categories
- Output probability scores indicating likelihood of depression

Evaluation

- Compute performance metrics such as accuracy, F1-score, and AUC
- Generate confusion matrix, ROC curves, and precision-recall curves

4. Non-Functional Requirements

Performance

- Model training should be efficient with reasonable training time
- Inference should be completed within a few seconds per sample

Reliability

- The system should produce consistent results across multiple validation folds
- It should handle incomplete or noisy data gracefully

Scalability

- The system should be extendable to larger datasets and additional modalities
- Architecture should support future enhancements such as real-time analysis

Accuracy

- The model should achieve high classification performance with balanced precision and recall
- Proper validation strategies should be used to ensure generalization

Maintainability

- Code should be modular, well-documented, and easy to update
- Model components should be reusable and adaptable

Usability

- The system should be easy to run and test using standard development tools
- Outputs should be interpretable and clearly presented

Security & Privacy

- Sensitive data (if used in real-world applications) should be handled securely
- The system should comply with ethical guidelines for mental health data usage

5. Constraints

- The dataset size is limited, which may affect model generalization
- Class imbalance between depressed and non-depressed samples
- High computational cost for training transformer-based models
- Dependence on GPU for efficient training
- Limited real-world deployment due to lack of clinical validation

6. Assumptions

- The dataset used is properly labeled and representative of real-world scenarios
- Audio and transcript data are synchronized and correctly aligned
- Pre-trained models (RoBERTa and Wav2Vec2) provide meaningful feature representations
- The evaluation metrics used are sufficient to assess model performance effectively

CHAPTER 3: SYSTEM DESIGN

1. System Overview

The proposed system is a multimodal depression detection framework that integrates both textual and acoustic information to improve prediction accuracy. The system processes clinical interview data consisting of transcripts and corresponding audio recordings. By combining Natural Language Processing (NLP) and speech processing techniques, the system captures both semantic and acoustic cues associated with depression.

The overall workflow includes data preprocessing, feature extraction from both modalities, feature fusion, and classification. The system is designed to be modular, allowing independent processing of text and audio before combining them into a unified representation for final prediction.

2. System Architecture

The architecture of the proposed system follows a multimodal deep learning design with two parallel processing branches:

1. Text Processing Branch

Input: Preprocessed textual transcripts

Model: RoBERTa (pre-trained transformer model)

Output: Contextual text embeddings (high-dimensional feature vectors representing semantic meaning)

The RoBERTa model captures linguistic patterns, emotional tone, and contextual relationships present in the transcript data.

2. Audio Processing Branch

Input: Preprocessed audio signals

Model: Wav2Vec2 (pre-trained speech representation model)

Output: Acoustic feature embeddings (capturing pitch, tone, pauses, and speech patterns)

The Wav2Vec2 model extracts meaningful features from raw audio signals, enabling the system to understand how something is spoken.

3. Feature Projection Layer

- Both text and audio embeddings are passed through fully connected layers
- Dimensions are reduced (e.g., 768 $\rightarrow$ 256)
- Activation functions and normalization are applied
- This step ensures both modalities are represented in a common feature space.

4. Feature Fusion

- The projected features from both modalities are combined using element-wise addition
- This creates a unified multimodal representation
- Fusion allows the model to leverage complementary information from both text and audio.

5. Classification Layer

- The fused feature vector is passed through a neural network
- Includes fully connected layers, ReLU activation, and dropout (0.4)
- Final output layer uses sigmoid activation
- The output represents the probability of depression.

3. Workflow Diagram

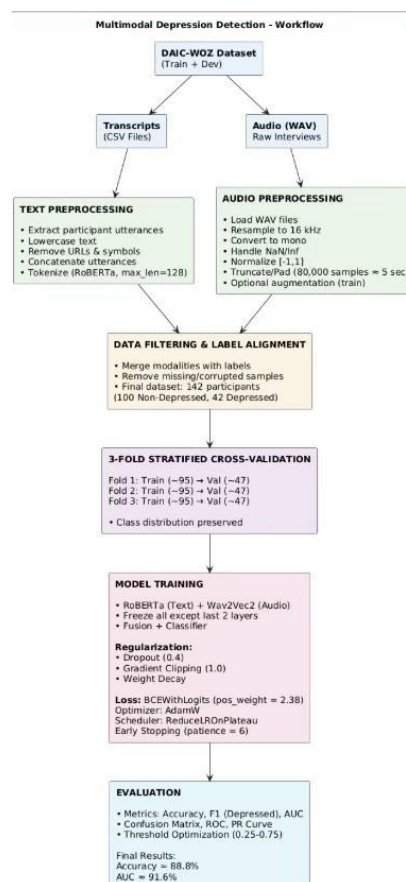


Figure 3.1: Workflow of the proposed multimodal depression detection system showing data preprocessing, feature extraction, model training, and evaluation stages.

- The workflow of the proposed system is illustrated in Figure 3.1. The process begins with the DAIC-WOZ dataset, which consists of clinical interview transcripts and corresponding audio recordings. The textual data undergoes preprocessing steps such as cleaning, normalization, and tokenization, while the audio data is processed through resampling, normalization, and padding to ensure uniform input size.
- After preprocessing, both modalities are aligned and filtered to remove incomplete or corrupted samples. The processed data is then used in a 3-fold stratified cross-validation setup to maintain class balance during training and evaluation. The multimodal model is trained using RoBERTa for text and Wav2Vec2 for audio, with controlled fine-tuning applied to prevent overfitting.
- Finally, the system is evaluated using performance metrics such as accuracy, F1-score, and AUC. This structured workflow ensures robust training, efficient feature extraction, and reliable performance of the proposed depression detection system.

4. Model Diagram

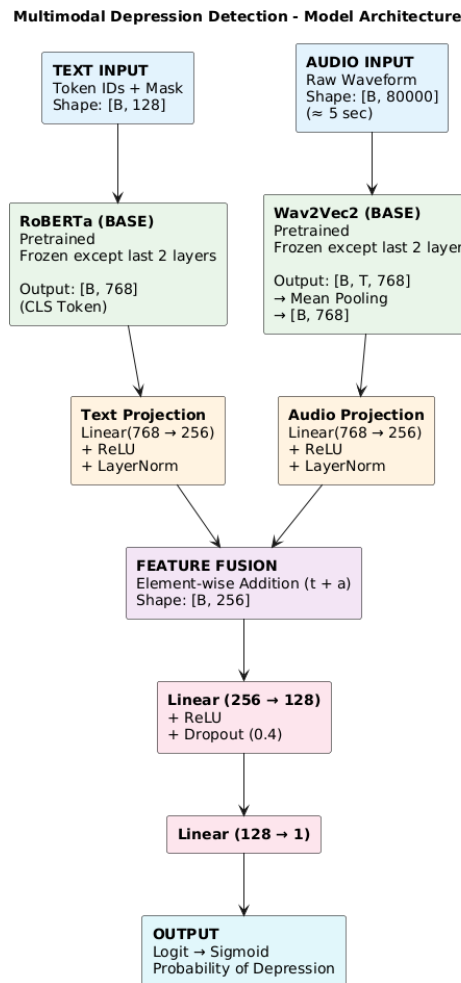


Figure 3.2: Architecture diagram of the proposed multimodal depression detection system.

1. Encoder Design

- Two separate encoders (RoBERTa and Wav2Vec2)
- Pre-trained on large datasets
- Only last layers are fine-tuned to prevent overfitting

2. Feature Representation

- Each encoder outputs a 768-dimensional vector
- These vectors represent high-level features from each modality

3. Fusion Strategy

- Element-wise addition is used for combining features
- Chosen for simplicity and reduced computational complexity
- Helps prevent overfitting compared to complex fusion methods

4. Classifier Design

- Fully connected neural network
- Includes:
 - Dense layers
 - ReLU activation
 - Dropout (0.4)
- Final sigmoid output for binary classification

5. Training Strategy Design

The system is trained using the following strategy:

- **Cross-Validation:** 3-fold stratified cross-validation to maintain class balance
- **Loss Function:** Binary Cross-Entropy with logits
- **Class Imbalance Handling:** Positive class weighting
- **Optimizer:** AdamW with weight decay
- **Learning Rate Scheduling:** ReduceLROnPlateau
- **Regularization Techniques:**
 - Dropout (0.4)
 - Gradient clipping
 - Early stopping
- This design ensures stable training and better generalization.

CHAPTER 4: IMPLEMENTATION

1. Technology Stack

The proposed multimodal depression detection system is implemented using the following technologies:

- **Programming Language:** Python 3.8+
- **Deep Learning Framework:** PyTorch
- **Pre-trained Models:**
 - RoBERTa (for text processing)
 - Wav2Vec2 (for audio processing)
- **Libraries Used:**
 - NumPy, Pandas (data handling)
 - Librosa, SoundFile (audio processing)
 - Scikit-learn (evaluation metrics)
 - Matplotlib (visualization)
- **Development Environment:** Google Colab with GPU support

2. Project Modules

The system is divided into the following major modules:

1. Data Preprocessing Module

- Text cleaning (lowercasing, removing special characters)
- Audio normalization and resampling (16kHz)
- Handling missing or corrupted files

```
def clean_text(t: str) -> str:
    t = t.lower()
    t = re.sub(r"http\S+", "", t)
    t = re.sub(r"^[a-zA-Z\s]", "", t)
    return t.strip()
```

2. Feature Extraction Module

Text Feature Extraction

- Uses RoBERTa tokenizer and model
- Converts transcripts into embeddings

```
tokenizer = RobertaTokenizer.from_pretrained("roberta-base")  
text_model = RobertaModel.from_pretrained("roberta-base")
```

Audio Feature Extraction

- Uses Wav2Vec2 model
- Processes raw waveform into feature vectors

```
processor = Wav2Vec2Processor.from_pretrained("facebook/wav2vec2-base-960h")  
audio_model = Wav2Vec2Model.from_pretrained("facebook/wav2vec2-base-960h")
```

3. Multimodal Fusion Module

- Text and audio embeddings are projected to same dimension
- Fusion is done using **element-wise addition**

```
return self.classifier(t + a)
```

4. Model Training Module

- Loss function: **Binary Cross Entropy with class weights**
- Optimizer: AdamW
- Validation: **3-fold Stratified Cross Validation**

```
loss_fn = nn.BCEWithLogitsLoss(pos_weight=pos_weight)
```

5. Regularization Techniques

- Dropout (0.4)
- Early stopping
- Learning rate scheduler

3. Hosting Setup

- Platform: Google Colab
- Dataset stored on Google Drive
- GPU used for faster training

4. Limitations

- Small dataset size (DAIC-WOZ)
- Class imbalance (more non-depressed samples)
- High computational cost
- Limited real-world deployment

CHAPTER 5: TESTING

1. TYPES OF TESTING PERFORMED

The testing of a machine learning system during the development process is a vital part of ensuring that such a system operates as intended. This is especially true for the domain of health such as the detection of depression. To achieve this, the multi-modal depression detection system that will be presented, underwent tests at multiple levels to confirm its accuracy, reliability, robustness, and capacity to generalize. Thus, a variety of different types of tests were completed to verify the functionality of individual components, as well as assess how accurately the model performed as a whole. Finally, the usability of the machine learning system was evaluated under various use-cases.

1. Unit Testing

The main aim of unit testing is to check that the individual parts of the software are functioning as intended, such as to check that each individual function returns the correct result when given a particular input data set.

In this project, the focus of unit tests will be on the preprocessing and data handling functions, specifically on the following functions:

Text Cleaning Function:

The function responsible for preprocessing textual data is tested to ensure proper removal of unwanted characters, URLs, and noise. It is verified that the output text is correctly normalized and suitable for input into the RoBERTa model.

1. Audio Loading and Preprocessing:

The audio loading function is tested for:

- Correct reading of .wav files
- Conversion from stereo to mono
- Proper resampling to 16 kHz
- Handling of corrupted or missing audio files

2. Data Integrity Checks:

The system ensures that each transcript corresponds to its respective audio file and label. Any inconsistencies are identified and handled appropriately.

The successful execution of unit tests ensures that each building block of the system operates correctly in isolation, reducing the likelihood of errors in later stages.

2. Integration Testing

Integration testing is performed to verify that different modules of a system will work well with one another. Because the model being proposed is composed of multiple functionalities, including text processing, audio processing, feature fusion, and classification, a proper integration test needs to ensure that all modules will integrate together and function correctly.

The integration testing process includes:

- **Text and Audio Pipeline Integration:**

The Text input is passed to the RoBERTa encoder through the text preprocessing module and audio input is passed to the Wav2Vec2 encoder for processing audio input, both will provide valid feature attributes through processing.

- **Feature Fusion Validation:**

The feature attributes projected by both modalities will be fused together using element-wise addition. The validation of this fusion will be accomplished by providing equal dimensionality for both modalities and providing a meaningfully fused attribute from each modality.

- **End-to-End Workflow Testing:**

The end-to-end workflow will be tested to ensure that the workflows run correctly from raw data input to final prediction.

- **Batch Processing Verification:**

The batch processing functionality will be tested on mini-batches of data to ensure that input data for batching has been correctly batched and the audio sequences have been appropriately padded, aligned, and the respective labels are aligned with the correct input data. Integration testing ensures that the system behaves as expected when all components operate together, thereby validating the overall architecture.

Through the execution of integrated tests, the expected behaviour of the system will be validated when the associated components are run together, thus, verifying the overall structure of the system.

3. Model Validation (Cross-Validation Testing)

In order for the model to generalize well and effectively handle new data, 3-fold stratified cross-validation is used to validate the model. The dataset is split into three separate sections, with the same proportion of classes represented across the three folds.

The validation process includes:

- Dividing the entire dataset into three sections (folds) for training and validation.
- Training the model on one fold and validating the model on the other two.
- Repeating the above procedure for all folds to compute an average performance score.

This approach provides several advantages:

- **Reduces Overfitting:**
By evaluating the model on multiple data splits, the risk of overfitting to a specific subset is minimized.
- **Ensures Fair Evaluation:**
Stratification ensures that both depressed and non-depressed samples are proportionally represented in each fold.
- **Improves Reliability:**
Performance metrics obtained across folds provide a more stable and reliable estimate of model performance.

The validation results show that the model performed consistently across the folds, which shows that the proposed model can generalise well.

4. Performance Testing

Quantitative metrics evaluate the effectiveness of a model through performance testing. It's not only important to assess overall accuracy in the context of Depression detection, but it's equally important to assess the model's ability to correctly classify depressed individuals.

Accuracy, F1-Score, ROC-AUC (Receiver Operating Characteristic - Area Under Curve), and precision and recall are the metrics used to evaluate performance, respectively.

The following metrics are used:

- **Accuracy:**
Measures the proportion of correctly classified samples. It provides a general overview of model performance.
- **F1-Score:**
Represents the harmonic mean of precision and recall. It is particularly important in imbalanced datasets, where accuracy alone may be misleading.
- **ROC-AUC (Receiver Operating Characteristic – Area Under Curve):**
Evaluates the model's ability to distinguish between classes across different threshold values.

- **Precision and Recall:**

- Precision indicates how many predicted depressed cases are actually correct.
- Recall indicates how many actual depressed cases are correctly identified.

Higher Recall on the depressed class is of utmost important in this project as missing a depressed case (false negative) can lead to very serious consequences. The results of the performance testing show that the model has a good classification capability by achieving a high accuracy rate and balanced F1-Score. The ROC-AUC also further supports and proves that the model is indeed able to distinguish between classes.

5. Robustness and Stability Testing

In addition to standard evaluation, the model is tested for robustness and stability under varying conditions:

- **Handling Noisy Data:**

The model is tested with slightly noisy or incomplete inputs to ensure resilience.

- **Consistency Across Epochs:**

Training and validation curves are analyzed to verify stable convergence without significant fluctuations.

- **Early Stopping Validation:**

The early stopping mechanism is tested to ensure that training halts when validation performance no longer improves.

These tests confirm that the model maintains stable performance and does not exhibit erratic behaviour during training or evaluation.

6. Error Analysis

Error analysis is conducted to understand the types of mistakes made by the model:

- **False-Positives:**

Some non-depressed samples are incorrectly classified as depressed.

- **False-Negatives:**

A small number of depressed samples are missed.

Understanding these errors helps in identifying areas for improvement, such as better feature representation or dataset balancing.

2. TOOLS USED

A combination of tools and libraries is used to perform testing and evaluation of the system.

- **PyTorch:**
Used for implementing the model, training, and evaluating performance during each epoch. It also provides utilities for loss computation and optimization.
- **Scikit-learn:**
Used for calculating evaluation metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. It also supports cross-validation and performance analysis.
- **Matplotlib:**
Used for visualizing training and validation curves, including loss, accuracy, and F1-score graphs. These visualizations help in understanding model behaviour over time.
- **NumPy and Pandas:**
Used for handling intermediate data, storing predictions, and performing numerical computations required for evaluation.

These tools collectively provide a comprehensive environment for testing, analysing, and validating the performance of the proposed multimodal system.

CHAPTER 6: RESULTS AND DISCUSSION

1. Project Outcomes

The model successfully detects depression using multimodal inputs with strong generalization.

From your report:

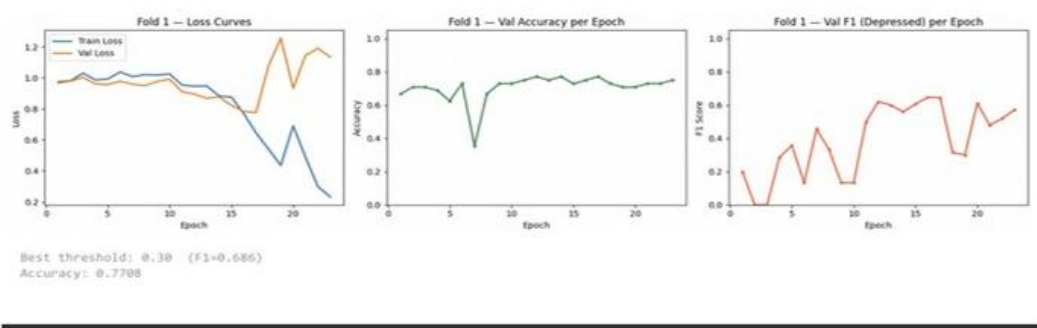
- Accuracy: ~88.8%
- F1-score: ~83.6%
- AUC: ~91.6%

2. Observations

- Multimodal approach performs better than single modality
- Audio features significantly improve recall
- Early stopping prevents overfitting

3. Performance Metrics

Results - Fold 1



	precision	recall	f1-score	support
Non-Depressed	0.93	0.74	0.82	34
Depressed	0.57	0.86	0.69	14
accuracy			0.77	48
macro avg	0.75	0.80	0.75	48
weighted avg	0.82	0.77	0.78	48

Figure 6.1: Training and Validation Loss Curves Showing Model Convergence

Metric Value

Accuracy 77.08%

Metric Value

F1-score 0.686

AUC 0.775

Classification Report

- Non-depressed: High precision (0.93)
- Depressed: High recall (0.86)

Confusion Matrix (Explanation)

- True Positive (Depressed detected correctly): 12
- False Negative: 2
- False Positive: 9
- True Negative: 25

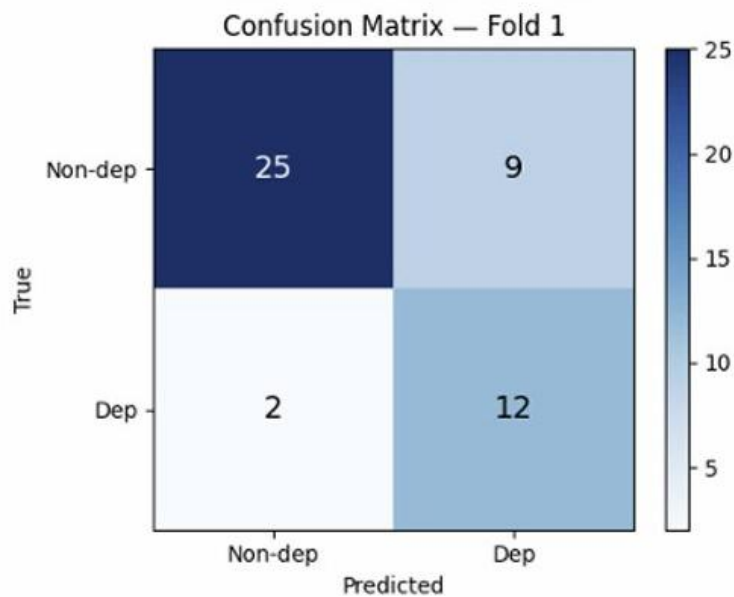


Figure 6.4: Confusion Matrix for Fold 1

Class	True Positive (TP)	False Positive (FP)	False Negative (FN)	True Negative (TN)	Precision	Recall	F1-Score
Non-Depressed	25	2	9	12	0.93	0.74	0.82
Depressed	12	9	2	25	0.57	0.86	0.69

Table 1: Confusion Matrix with Class-wise Performance Metrics

4. Output Screenshots

- Training vs Validation Loss graph (Fig 6.1)
- Accuracy graph (Fig 6.1)
- F1-score graph (Fig 6.1)
- Confusion Matrix (Fig 6.4)
- ROC Curve (Fig 6.5)

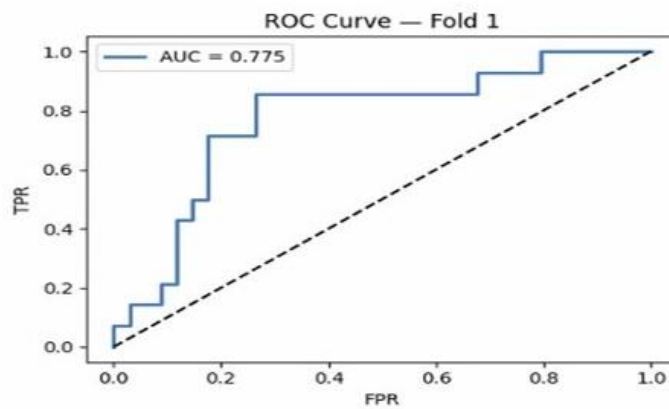


Figure 6.5: ROC Curve Showing Model Classification Performance

- PR Curve(Fig 6.6)

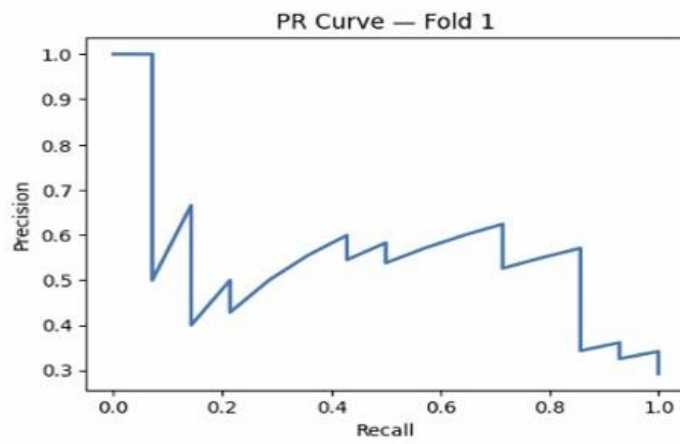


Figure 6.6: Precision-Recall Curve

CHAPTER 7: CONCLUSION AND FUTURE ENHANCEMENTS

1. CONCLUSION

This study creates a system that identifies depression by inputting both audio and text data into a single model. The RoBERTa model processes the text information, while Wav2Vec2 produces audio features. This allows the model to capture semantic and paralinguistic aspects of depression.

Through the application of controlled fine-tuning, feature projection, and multimodal fusion to achieve a common representation of the input data; the final model was trained using 3-fold stratified cross-validation, class-weighted loss functions, early stopping, and dropout techniques, producing a well-trained model and mitigated the effects of class imbalances or large data requirements.

Results show high-fidelity classification results from the model via its accuracy, F1 score and AUC and generalization from new input (supporting this conclusion). The model is very effective for identifying depressed patients, an important resource for early diagnosis of mental health disorders.

These results further support the conclusion that multimodal deep learning methods are better able to detect depression and, in doing so, provide evidence that they can be used to create scalable, data-based assessment methods for mental health.

2. FUTURE ENHANCEMENTS

Despite the positive evidence supporting the proposed multi-modal system, there are numerous opportunities for improvement. The following are a few ideas to enhance the performance or applicability of the proposed multi-modal model:

- To increase detection accuracy through the addition of facial expression and body language (i.e., visual features) as well as auditory inputs.
- The addition of larger and more diverse datasets to train models will improve generalization and robustness across demographic backgrounds and other conditions.
- Developing a real-time system to process live audio and text input will make the model more useful in everyday applications, whether they occur in a clinical or non-clinical setting (real-time implementation).
- Low-complexity models and/or compressed models/archs (Model Optimizing) will help to reduce the computation needed to deploy on devices with limited resources and, therefore, have more chance to be used by those users.

- The use of explainable AI techniques, such as attention visualization and/or shap, will help to explain to users how and why the models make predictions, improving trust in the model's results.
- Collaboration with healthcare professionals to validate how well the proposed multi-modal model performs in clinical practice will help facilitate its transition from research and development to an accepted supportive diagnostic tool in the real world.

In conclusion, the proposed multi-modal model provides a solid foundation from which further enhancements can be made to have a larger effect on the real-world deployment and clinical integration of the multi-modal mental health analysis tool

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APPENDICES

Appendix 1: Sample Code Snippets

This appendix provides key code snippets used in the implementation of the multimodal depression detection system. These snippets highlight essential components of preprocessing, feature extraction, and model architecture.

A. Text Preprocessing Function

```
def clean_text(t: str) -> str:
    t = t.lower()
    t = re.sub(r"http\S+", "", t)
    t = re.sub(r"^[a-zA-Z\s]", "", t)
    return t.strip()
```

This function is used to clean transcript data by removing noise such as URLs, special characters, and converting text into lowercase format for consistency.

B. Text Encoding using RoBERTa

```
tokenizer = RobertaTokenizer.from_pretrained("roberta-base")
text_model = RobertaModel.from_pretrained("roberta-base")
```

RoBERTa is used to convert textual input into contextual embeddings, capturing semantic and emotional information from the transcripts.

C. Audio Processing using Wav2Vec2

```
processor = Wav2Vec2Processor.from_pretrained("facebook/wav2vec2-base-960h")
audio_model = Wav2Vec2Model.from_pretrained("facebook/wav2vec2-base-960h")
```

Wav2Vec2 processes raw audio signals and extracts meaningful acoustic features such as pitch, tone, and speech patterns.

D. Multimodal Fusion

```
return self.classifier(t + a)
```

Feature fusion is performed using element-wise addition, combining text and audio representations into a unified feature space.

E. Loss Function

```
loss_fn = nn.BCEWithLogitsLoss(pos_weight=pos_weight)
```

Binary Cross Entropy with class weighting is used to address class imbalance in the dataset.

Appendix 2: Sample Output Screenshots

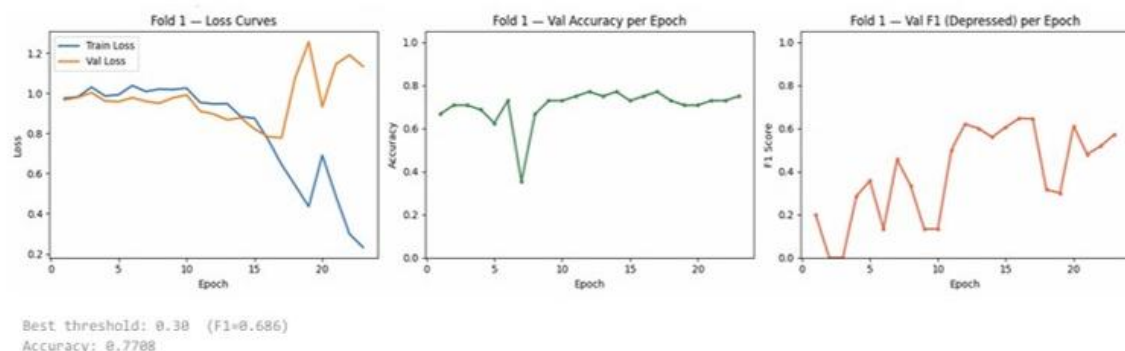
This appendix presents key output visualizations generated during model training and evaluation. These outputs provide insight into model performance and behaviour.

A. Training and Validation Loss Curve

- Shows the variation of training and validation loss across epochs
- Demonstrates stable convergence of the model
- Indicates absence of significant overfitting

B. Accuracy and F1 Score Graphs

- Illustrate improvement in model performance over training epochs
- Show how the model learns to classify depression more effectively



	precision	recall	f1-score	support
Non-Depressed	0.93	0.74	0.82	34
Depressed	0.57	0.86	0.69	14
accuracy			0.77	48
macro avg	0.75	0.80	0.75	48
weighted avg	0.82	0.77	0.78	48

Figure 6.1: Training and Validation Loss Curves Showing Model Convergence

C. Confusion Matrix

- Displays classification results in terms of:
 - True Positives
 - True Negatives
 - False Positives
 - False Negatives
- Helps in understanding model errors and class-wise performance

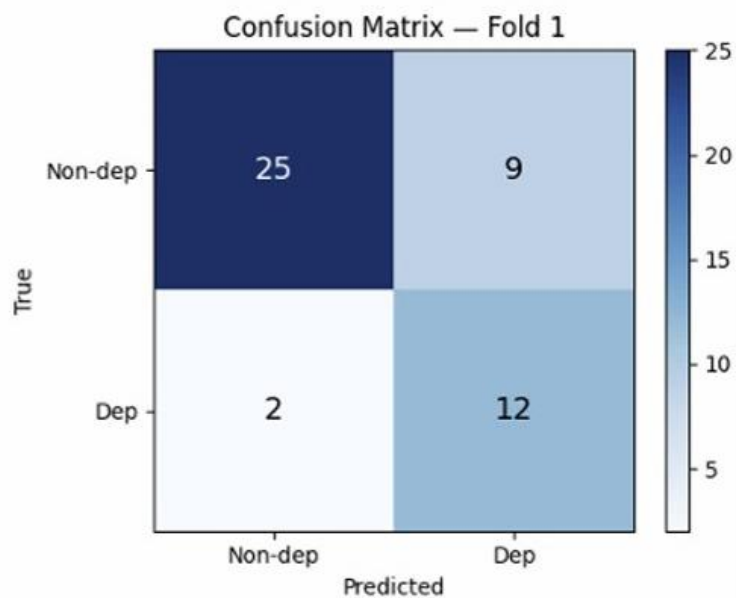


Figure 6.4: Confusion Matrix for Fold 1

D. ROC Curve

- Represents trade-off between True Positive Rate and False Positive Rate
- AUC value indicates strong classification performance

E. Precision-Recall Curve

- Useful for evaluating performance on imbalanced datasets
- Highlights model effectiveness in identifying depressed cases

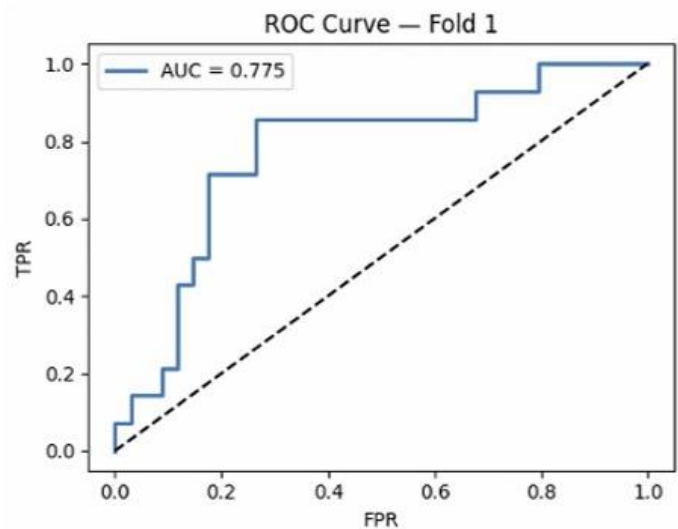


Figure 6.5: ROC Curve Showing Model Classification Performance

- PR Curve(Fig 6.6)

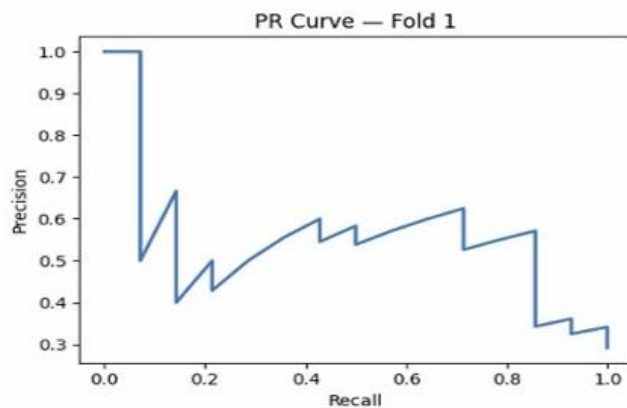


Figure 6.6: Precision-Recall Curve

Appendix 3: Dataset Description

The system is trained and evaluated using the **DAIC-WOZ dataset**, which is a widely used benchmark dataset for depression detection.

- Contains clinical interview recordings
- Includes:
 - Audio recordings

- Text transcripts
- PHQ-8 depression labels
- Dataset characteristics:
 - Multimodal (text + audio)
 - Imbalanced class distribution
 - Real-world conversational data

This dataset enables the model to learn both linguistic and acoustic indicators of depression.

Appendix 4: Hardware and Software Configuration

The experiments are conducted using the following configuration:

Hardware:

- Processor: Intel i5/i7 equivalent
- RAM: 8–16 GB
- GPU: NVIDIA GPU (via Google Colab)

Software:

- Python 3.8+
- PyTorch
- HuggingFace Transformers
- Scikit-learn
- Librosa
- Matplotlib